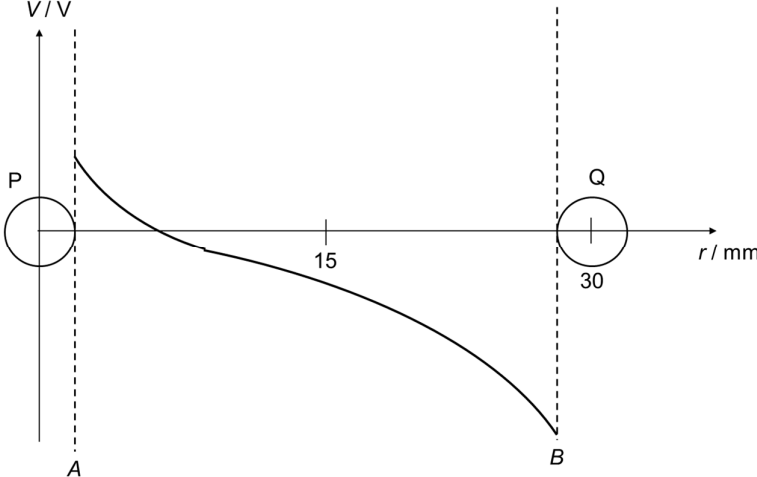


5 (a)	Electric potential at a point is the work done per unit positive charge to bring a small charge from infinity to that point	B1
(b)(i)	 [B1] shape: non-zero gradient [B1] r axis intercept nearer to P [B1] magnitude of V at B > magnitude of V at A	
(b)(ii)1	Electric potential energy = $\frac{Q_P Q_Q}{4\pi\epsilon_0 r}$ $(9 \times 10^9) \frac{(+20 \times 10^{-6})(-40 \times 10^{-6})}{0.030}$ $= -240 \text{ J}$	C1
(b)(ii)2	By conservation of energy, $KE_i + EPE_i = KE_f + EPE_f$ $\frac{1}{2}(0.20 \times 10^{-3})u^2 + (-240) = 0$	A1 C1
	$u = 1550 \text{ m s}^{-1}$	A1

6 (a)	Max KE possessed by electron emitted from surface level	B1
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